Which Class Did Better?

AP Statistics · Unit 1 · Topic 1.9 · B: 2026-09-24 · E: 2026-09-25 · TEACHER KEY

CED TETHER

- **Skill 2.D** — Compare distributions or relative positions of points within a distribution.
- **Enduring Understanding UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.N** — Compare graphical representations for multiple sets of quantitative data. [Skill 2.D]
- **EK UNC-1.N.1** — Any of the graphical representations, e.g., histograms, side-by-side boxplots, etc., can be used to compare two or more independent samples on center, variability, clusters, gaps, outliers, and other features.
- **LO UNC-1.O** — Compare summary statistics for multiple sets of quantitative data. [Skill 2.D]
- **EK UNC-1.O.1** — Any of the numerical summaries (e.g., mean, standard deviation, relative frequency, etc.) can be used to compare two or more independent samples.

Essential question: What does 'did better' mean for a whole class, and which feature of a plot answers it?

DOK-3 skill: 4.B · **FRQ pattern:** `compare-two-boxplots-adjudicate-competing-claims-then-choose-one-statistic`

DOK rationale: Part (c) requires weighing two competing claims about which class did better against several features of two boxplots at once, then committing to a single statistic and defending it; either claim can be supported by one feature alone, so the answer depends on coordinating center, variability, and an outlier rather than on any single procedure.

First take → rules + (a)–(c) → turn in **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Read Sam's and Tia's lines aloud once; take a quick hand-count of first takes but do not react.
- At minute 5, read the rules box; have students start (a).
- Board slide back up at minute 33. Circulate with the prompts; insist on comparative sentences.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- One-sentence first take naming one plot feature.
- Use the rules box to compare the classes in (a) and (b), with points and comparative words.
- Finish (a)–(c); exit line; turn in.

Questions to ask

- How many students does the top of a whisker represent? How many does the median line represent?
- If the 40 in Period B were a 33 instead, would Sam's claim change? Would Tia's?
- Point at the box widths. Which class would you rather be a random student in? Why?
- Your parent asks 'how did the class do?' — which number answers that question?

Adult role
Solo teacher. Circulate 5 pairs. Answer questions with questions; do not confirm Tia until two features are cited.

[WARN] LOOK FOR

- Sentences that describe one plot at a time with no comparison word.
- Reading the whisker end as 'most students'.
- Choosing the mean for Period B without noticing the outlier at 5.

SCORING PART (c) — E / P / I

Expected elements

- **tia-center** (required) — Chooses Tia using medians 31 versus 27
- **compares-spread** (required) — Uses IQR 8 versus 11, overall ranges 20 versus 35, or B's low outlier as further evidence
- **median-over-max** (required) — Chooses median and explains a maximum describes an extreme observation rather than the class center

E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

Common mistakes

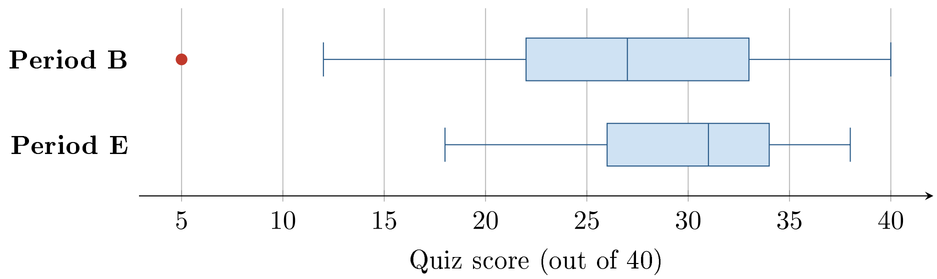
- Reading the maximum as 'the class did better'
- Describing each boxplot separately instead of comparing (no comparative language)
- Calling the wider box 'more scores' rather than 'more variable'
- Choosing the mean for Period B without noticing the outlier at 5 pulls it down

Optional #1 — not collected
DOK 2 Two bus routes were timed for 20 mornings each. Route 1: mean 24.0 min, SD 6.5 min, median 22 min. Route 2: mean 24.5 min, SD 1.8 min, median 24 min. Write two sentences comparing the routes for a student who must not be late, using comparative language and the statistics given.

Key: The means are nearly the same (24.0 versus 24.5 minutes), while Route 2 has much lower SD (1.8 versus 6.5 minutes), so it is more consistent. A student concerned about unpredictable travel could prefer Route 2, but these summaries alone do not give the worst delay or the probability of being late.

[STAR] TODAY'S PROBLEM — annotated

Two AP Statistics classes took the same 40-point quiz. The side-by-side boxplots show the scores. Two students read them differently. **Sam:** "Period B did better — its highest score is 40 and Period E's is only 38." **Tia:** "Period E did better — its median is higher and its scores are more consistent."



FIRST TAKE (5 min)

Which class did better on the quiz? One sentence, and name ONE feature of the plot you used.
Key: Not graded. Expect a split toward Sam (the 40 is eye-catching); that split is the lesson.

Rules callout "COMPARING DISTRIBUTIONS (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Read the median for each class.

Key: Period B: 27 points. Period E: 31 points.

DOK 2 **(b)** In two or three sentences, compare the quiz scores' center, spread, apparent shape, and unusual features. Use points and comparative words.

Key: E has a higher median (31 versus 27 points) and a smaller IQR (8 versus 11 points). B's overall range is $40 - 5 = 35$ points, versus E's $38 - 18 = 20$, and B has the low outlier at 5. B is fairly balanced within its box but extends farther on the low side overall; E also appears more spread toward low scores. Boxplots cannot establish detailed shape.

DOK 3 **(c)** Whose claim is better supported, Sam's or Tia's? Cite two specific features of the boxplots that settle it. Then: if you could report only ONE statistic per class to a parent asking 'how did the class do?', which statistic would you choose, and why that one rather than the maximum?

Key: Tia's. Two settling features: the medians (31 vs. 27 — the typical student in E scored higher) and the spread (E's IQR 8 vs. B's 11, plus B's outlier at 5). Sam's evidence is a single value: one student in B scored 40, which says nothing about the class. Report the median: it describes the typical student and is less affected by extreme scores; the maximum describes exactly one student and would make Period B look better because of one paper.

EXIT

One thing the rules box changed about my first take: ______________________________